Universitatea
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FACULTATEA DE INGINERIE ELECTRICĂ
ȘI ȘTIINȚA CALCULATOARELOR

Proiect Inginerie Audio

Sistem analogic de prelucrare a semnalelor de audiofrecvență

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4LF632 - TSTC

Date de proiectare

Nr. crt	Rin,min [M□]	G_1 [dB]	G2NAB [dB]	G2RIAA [dB]	Uies,mix [V]	CORECTOR DE TON	
						f_L[Hz]	f_H[kHz]
11.	400	10	60	-	0.7	40	14

Etapa 1

- Date de proiectare

  Pentru preamplificatorul cu rezistență mare de intrare:

  - $U_{ies} = 0.7V$
  - $G_1 = 10dB$
  - $R_{in,min} = 400M\Omega = 400 * 10^6\Omega$
  - $f_s = 20kHz = 20 * 10^3 Hz$
  - $SR_{calc} = 2\pi f_s U_{ies}$
  - $a_o = A_{VD} = 200\,V/mV = 200000\,V/V$
  - $f_u = B_1 = 3MHz = 3 * 10^6 Hz$
  - $r_d = r_i = 10^{12}\Omega$

- Criteriul SR

$$U_{ies} = 10 * U_{ies,mix} = 10 * 0.7V = 7V$$
$$\mathrm{SR}_{calc} = 2\pi f_s U_{ies} = 2 * 3.14 * 20 * 10^3 Hz * 7V = 879200\, V/s = 0.879\, V/us$$

$$SR_{AO} \geq SR_{calc} \;=>\; 8\, V/us \;\geq 0.879\;\; V/us$$

- <u>Criteriul rezistentei de intrare</u>

$$b_1 = \frac{1}{A_1} = \frac{1}{10^{\frac{G_1}{20}}} = \frac{1}{10^{\frac{10}{20}}} = \frac{1}{10^{\frac{1}{2}}} = 3.16 * 10^{-1}$$
$$f_a = \frac{f_u}{a_o} = \frac{3 * 10^6 Hz}{200 * 10^3\, V/V} = \frac{3 * 10Hz}{2} = 15Hz$$
$$f_A = b_1 * f_u = 3.16 * 10^{-1} * 3 * 10^6 Hz = 9.16 * 10^5 Hz$$
$$r_{d,calc} = \frac{R_{in,min}\sqrt{1+\left(\frac{f_S}{f_a}\right)^2}}{(1+a_0 \cdot b_1)\sqrt{1+\left(\frac{f_S}{f_A}\right)^2}} => \frac{4*10^8\Omega\sqrt{1+\left(\frac{20*10^3 Hz}{15Hz}\right)^2}}{(1+2*10^5*3.16*10^{-1})\sqrt{1+\left(\frac{20*10^3 Hz}{9.16*10^5 Hz}\right)^2}} => \frac{4*10^8\Omega*1333.3}{61601}$$
$$=> 8.657 * 10^6\Omega$$

Etapa 2 - Proiectarea preamplificatorului cu impedanță mare de intrare

Configuratia neinversoare

$A_1 = \sqrt{10}(\approx 3.16)$

$A_1 = 1 + \frac{R_{2-1}}{R_{1-1}}$ => $\frac{R_{2-1}}{R_{1-1}} = A_1 - 1$=>$\frac{R_{2-1}}{R_{1-1}} = \sqrt{10} - 1$

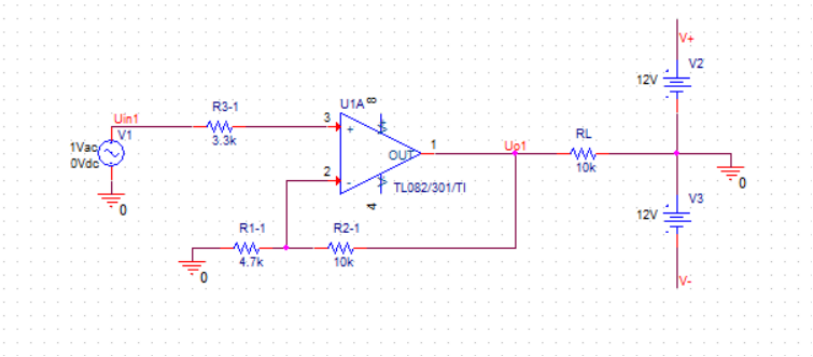
$R_{3-1} = R_{1-1}||R_{2-1}$

Alegem din seria E24 $R_{2-1} = 10k\Omega$

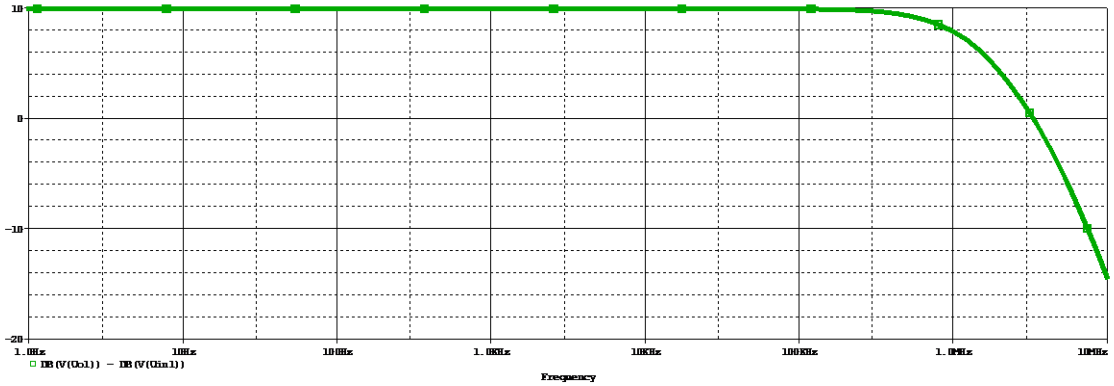
=>$R_{1-1} = \frac{R_{2-1}}{A_1 - 1} = \frac{10k}{\sqrt{10}-1} \approx 4624.75$ => Alegem din seria E24 - $R_{1-1} = 4.7k\Omega$

$R_{3-1} = \frac{R_{1-1} * R_{2-1}}{R_{1-1} + R_{2-1}} = \frac{10k * 4.7k}{10k + 4.7k} = \frac{47k}{14.7k} = 3.19$ =>Alegem din seria E24 - $R_{3-1} = 3.3k\Omega$

Schema circuit



Caracteristica DB(V(Uo1)) - DB(V(Uin1))

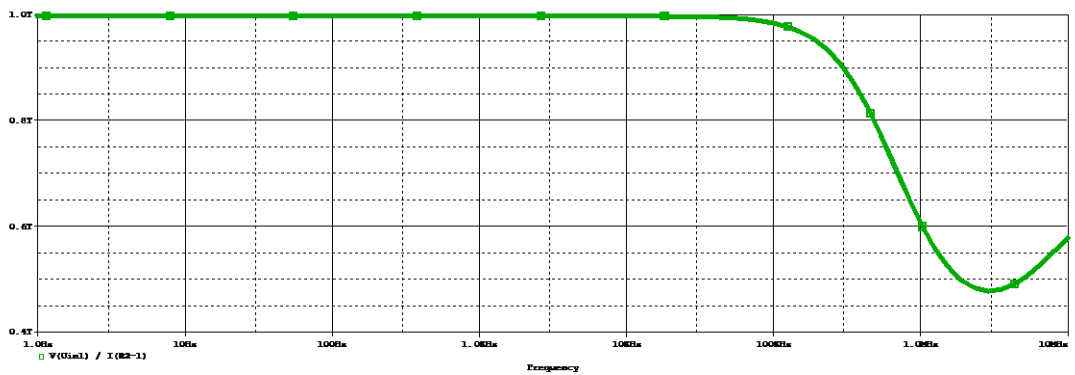


Fereastra Probe cursor - frecventa superioara

Probe Cursor

Trace Color	Trace Name	Y1	Y2	Y1 - Y2		Y1(Cursor1) - Y2(Cu	
	X Values	1.2773M	1.0000	1.2773M		Y1 - Y1(Cursor1)	Y2 - Y
CURSOR 1,2	DB(V(Uo1)) - DB(V(Uin1))	6.8860	9.904	-3.0180		0.000	0.000

Caracteristica V(Uin1) / I(R3-1)



Fereastra Probe cursor – $f_s = 20kHz$

Probe Cursor

	Trace Color	Trace Name	Y1	Y2	Y1 - Y2		Y1(Cursor1) - Y2(Cursor2	
		X Values	20.153K	1.0000	20.152K		Y1 - Y1(Cursor1)	Y2 - Y2(Curs
	CURSOR 1,2	V(Uin1) / I(R3-1)	0.9962T	0.9968T	-556.582M		0.000	0.000

Etapa 3 - Proiectarea preamplificatorului de bandă magnetică (corecție NAB)

Frecventele de frangere:

$$f_1 = 3183Hz$$
$$f_2 = 50Hz$$

Castig: $G_{2NAB} = 60dB$

$$f_1 = \frac{1}{2\pi R_{2-2} C_{2-2}}$$
$$f_2 = \frac{1}{2\pi 9(R_{2-2} + R_{3-2}) C_{2-2}}$$

Alegem $C_{2-2} = 10nF$

$R_{2-2} = \frac{1}{2\pi f_1 C_{2-2}} = \frac{1}{2\pi * 3183Hz * 10nF} = 5k$ =>Alegem din seria E24(5%) $R_{2-2} = 5.1k\Omega$

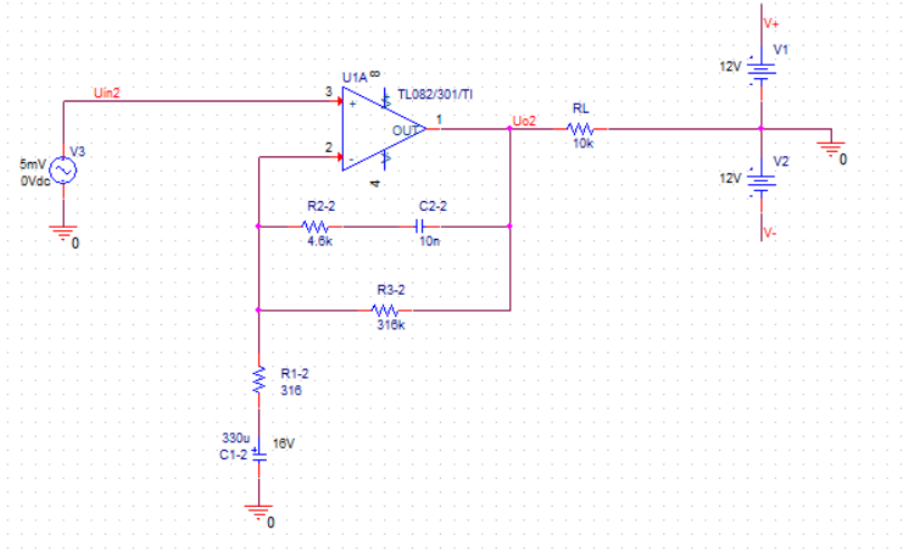
$R_{3-2} = \frac{1}{2\pi f_2 C_{2-2}} - R_{2-2} = \frac{1}{2\pi * 50Hz * 10nF} - 5.1k\Omega = 313.2k$ => Alegem din seria E96(1%)

$$R_{3-2} = 316k\Omega$$

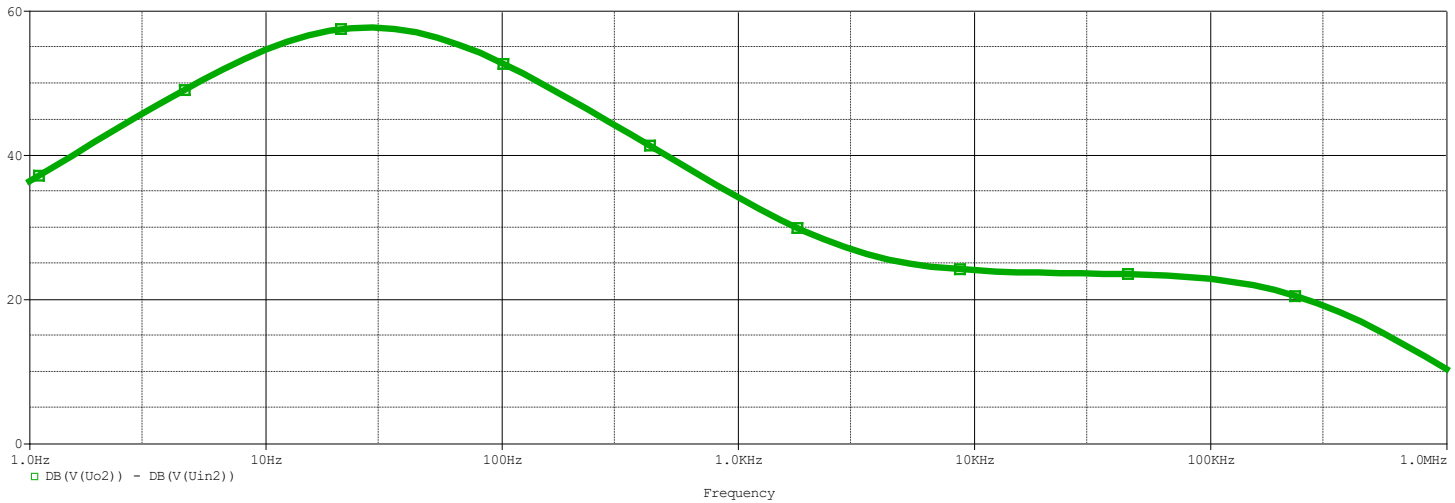
$$R_{1-2} = \frac{R_{3-2}}{10^{\frac{G_{2NAB}}{20}}-1} = \frac{316k}{10^3-1} = 316 \Rightarrow \text{Alegem din seria E96(1\%) } R_{1-2} = 316\Omega$$

$$C_{1-2} \geq \frac{1}{2\pi*20*R_{1-2}} \Rightarrow C_{1-2} \geq \frac{1}{2\pi*20*316} \Rightarrow C_{1-2} \geq 2.51*10^{-5} \Rightarrow \text{Alegem } C_{1-2} = 33uF$$

Schema circuitului



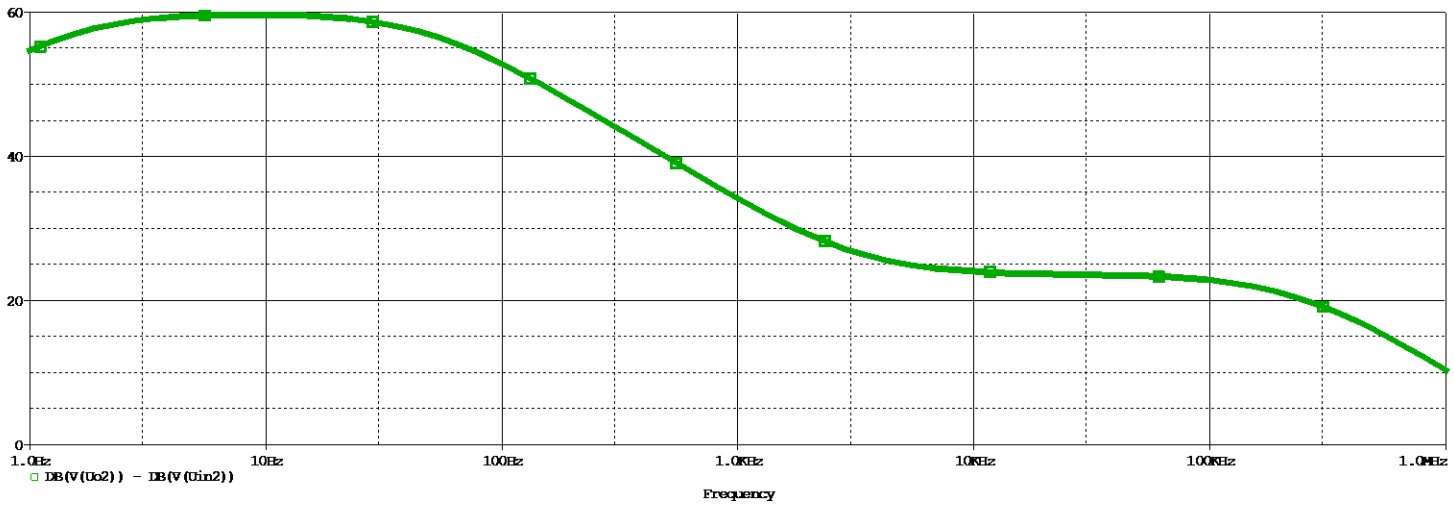
Caracteristica inainte de ajustare



Probe Cursor

	Trace Color	Trace Name	Y1	Y2	Y1 - Y2		Y1(Cursor1) - Y2(Curs	
		X Values	28.480	1.0000	27.480		Y1 - Y1(Cursor1)	Y2 - Y2(C
	CURSOR 1,2	DB(V(Uo2)) - DB(V(Uin2))	57.648	36.319	21.329		0.000	0.000

Caracteristica dupa ajustare $C_{1-2} = 330uF$



Probe Cursor

	Trace Color	Trace Name	Y1	Y2	Y1 - Y2		Y1(Cursor1) - Y2(Curs	
		X Values	17.308	1.0000	16.308		Y1 - Y1(Cursor1)	Y2 - Y2(C
	CURSOR 1,2	DB(V(Uo2)) - DB(V(Uin2))	59.422	54.772	4.6497		0.000	0.000

Fereastră pentru maxim (cursorul 1) și la -3dB (cursorul 2) pentru determinarea frecvenței f2;

Probe Cursor

Trace Color	Trace Name	Y1	Y2	Y1 - Y2		Y1(Cursor1) - Y2(Curs	
	X Values	17.308	55.158	-37.850		Y1 - Y1(Cursor1)	Y2 - Y2(
CURSOR 1,2	DB(V(Uo2)) - DB(V(Uin2))	59.422	56.417	3.0050		0.000	0.000

Fereastră pentru maxim (cursorul 1) și la -33dB (cursorul 2) pentru determinarea frecvenței f1;

Probe Cursor

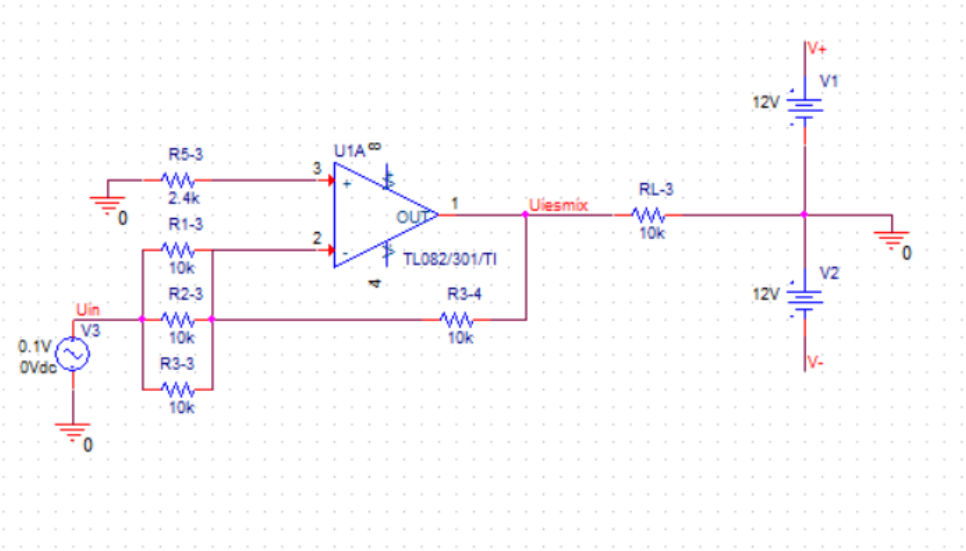
Trace Color	Trace Name	Y1	Y2	Y1 - Y2		Y1(Cursor1) - Y2(Cu	
	X Values	8.1113	3.1868K	-3.1787K		Y1 - Y1(Cursor1)	Y2 - Y
CURSOR 1,2	DB(V(Uo2)) - DB(V(Uin2))	59.706	26.697	33.009		0.000	0.000

Etapa 4 - Proiectarea mixerului analogic

$$R_{4-3} = R_{3-3} = R_{2-3} = R_{1-3} = R_{in,mix} = 10\text{k}\Omega$$

$$R_{5-3} = \frac{1}{\frac{1}{R_{1-3}}+\frac{1}{R_{2-3}}+\frac{1}{R_{3-3}}+\frac{1}{R_{4-3}}} = \frac{1}{\frac{1}{10\text{k}}+\frac{1}{10\text{k}}+\frac{1}{10k}+\frac{1}{10\text{k}}} = 2.5k\Omega$$ =>Alegem din seria E24(5%) $R_{5-3} = 2.4k\Omega$

Schema circuitului



Probe cursor – frecventa limita superioara

Probe Cursor

Trace Color	Trace Name	Y1	Y2	Y1 - Y2		Y1(Cursor1) - Y2
	X Values	954.175K	1.0000	954.174K		Y1 - Y1(Cursor1)
CURSOR 1,2	DB(V(Uiesmix)) - DB(V(Uin))	6.5568	9.542	-2.9855		0.000

Etapa 5 – Corector de ton

$f_{\mathrm{L}} = 40Hz$
$f_{\mathrm{H}} = 14kHz$
$R_{2-4} = 100k\Omega$

$R_{1-4} = \frac{R_{2-4}}{9}$ => $R_{1-4} = 11.11k\Omega$ =>Alegem din seria E96(1%) $R_{1-4} = 11.3k\Omega$
$R_{1\mathrm{a}-4} = R_{1\mathrm{b}-4} = R_{1-4}$ => $R_{1\mathrm{a}-4} = R_{1\mathrm{b}-4} = R_{1-4} = 11.3k\Omega$

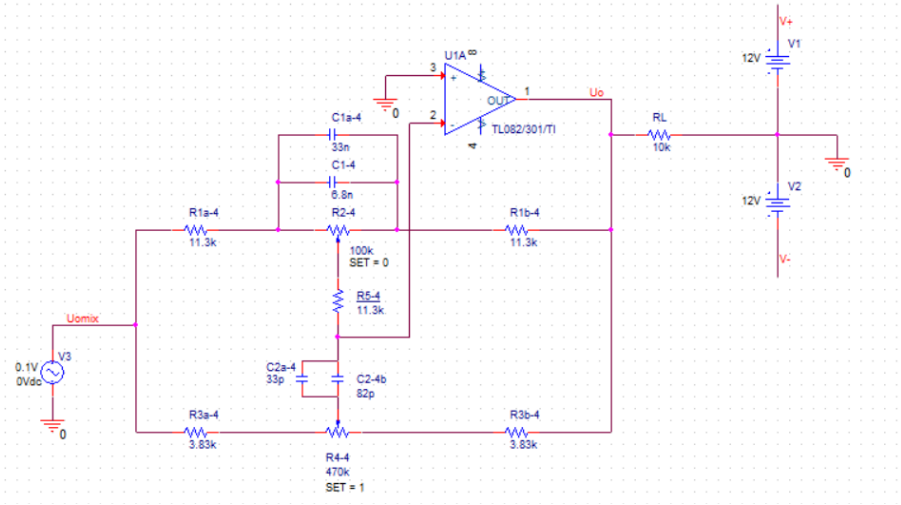
$R_{5-4} = R_{1-4}$ => $R_{5-4} = R_{1-4} = 11.3k\Omega$

$R_{3-4} = \frac{R_{1-4}+2*R_{5-4}}{9} = \frac{11.3k+2*11.3k}{9} = 3.76k\Omega$ => Alegem din seria E96(1%) $R_{3-4} = 3.83k\Omega$
$R_{3\mathrm{a}-4} = R_{3\mathrm{b}-4} = R_{3-4}$ => $R_{3\mathrm{a}-4} = R_{3\mathrm{b}-4} = R_{3-4} = 3.83k\Omega$
$C_{1-4} = \frac{1}{2\pi R_{2-4} f_{\mathrm{L}}} = \frac{1}{2*3.14*100k*40} = \frac{1}{25120000} \approx 3.98 * 10^{-8}F$
=> $C_{1-4} \approx 39.8nF$ => $C_{1\mathrm{a}-4} + C_{1\mathrm{b}-4} = C_{1-4}$ => $33nF + 6.8nF = 39.8nF$

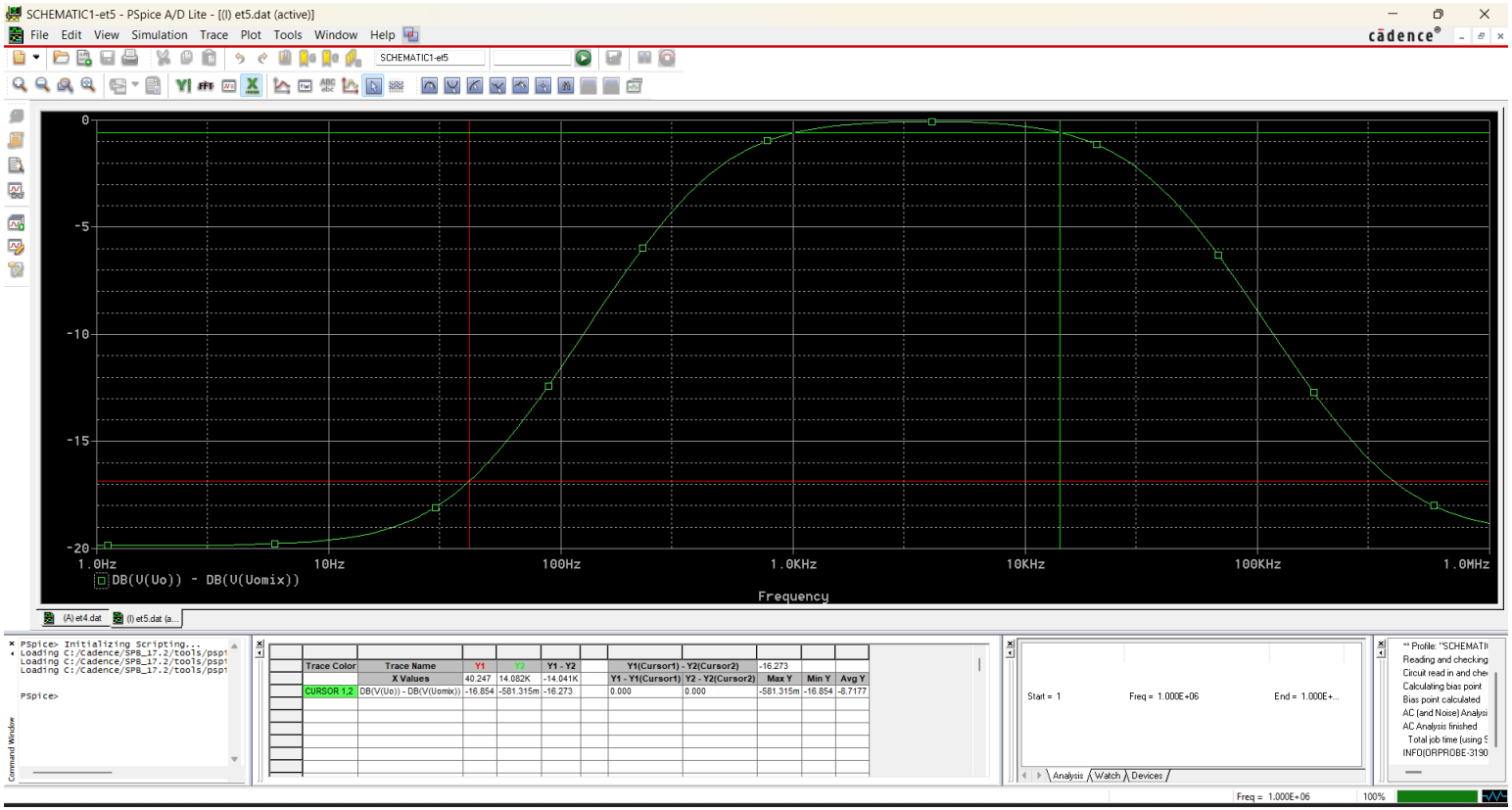
$$C_{2-4} = \frac{1}{2\pi R_{2-4} f_{\mathrm{H}}} = \frac{1}{2 * 3.14 * 100k * 14k} = \frac{1}{8792000000} \approx 1.14 * 10^{-10} F => C_{2-4} \approx 114pF$$

$$=> C_{2\mathrm{a}-4} + C_{2\mathrm{b}-4} = C_{2-4} => 82pF + 33pF = 115pF$$

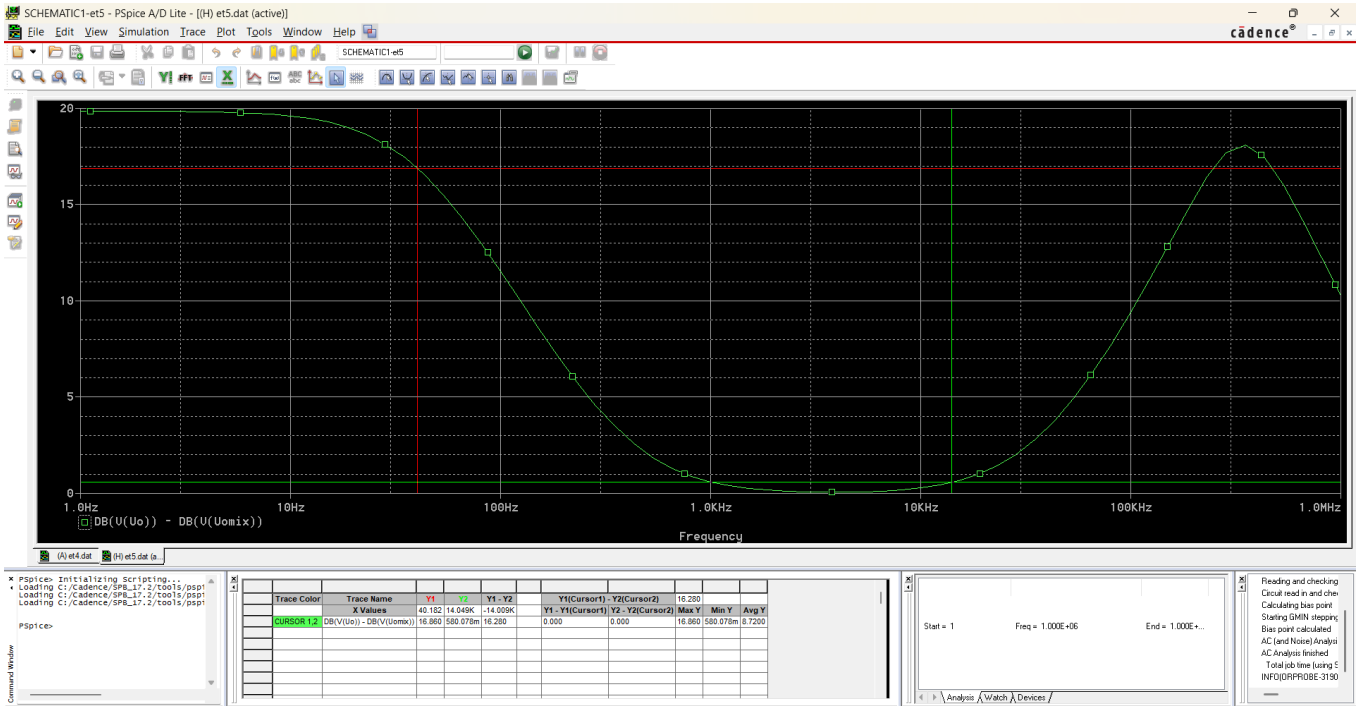
Schema circuitului



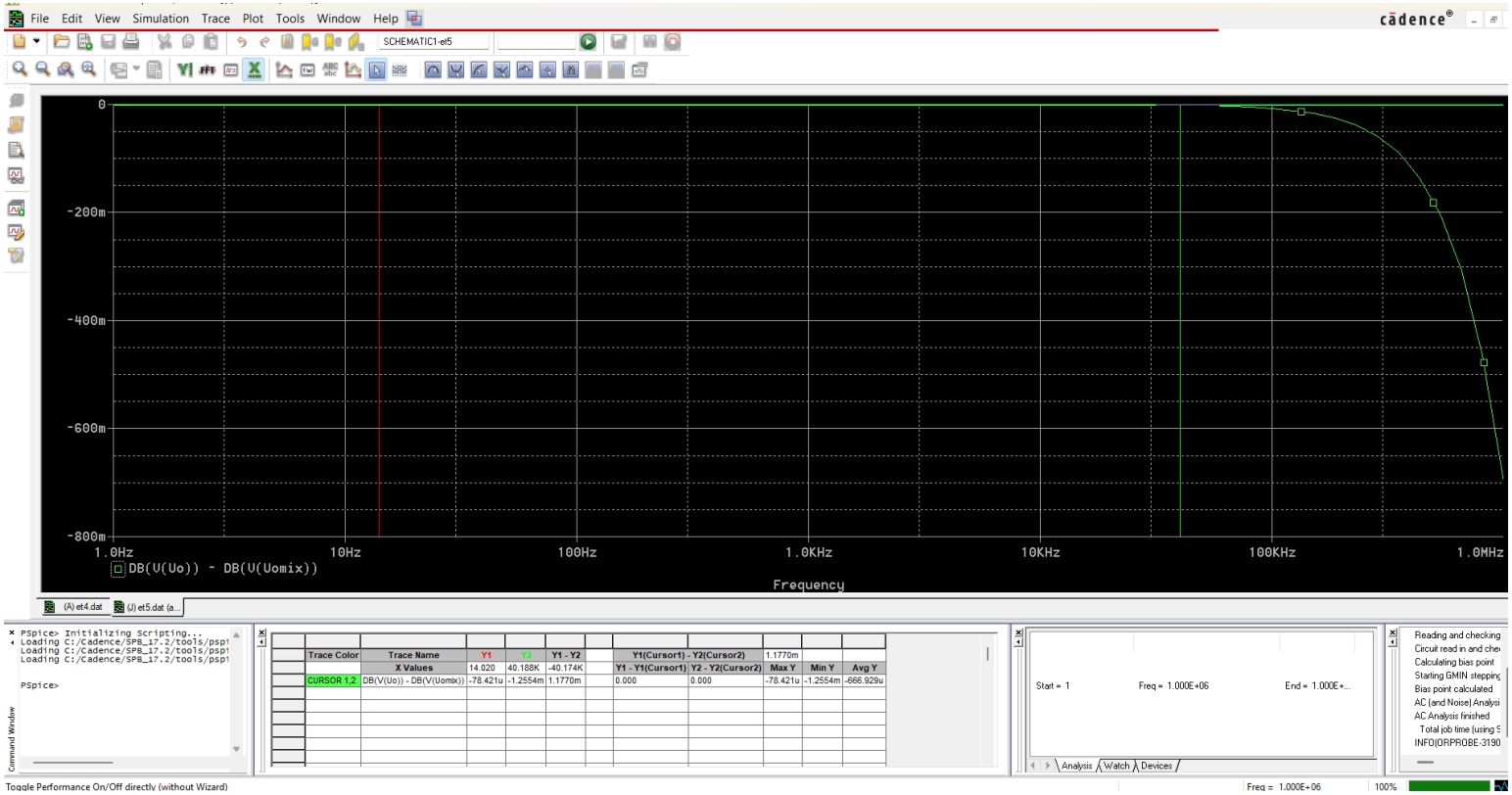
SET=0 la R2-4 și SET=1 la R4-4 – Atenuare



SET=1 la R2-4 și SET=0 la R4-4 - Amplificare

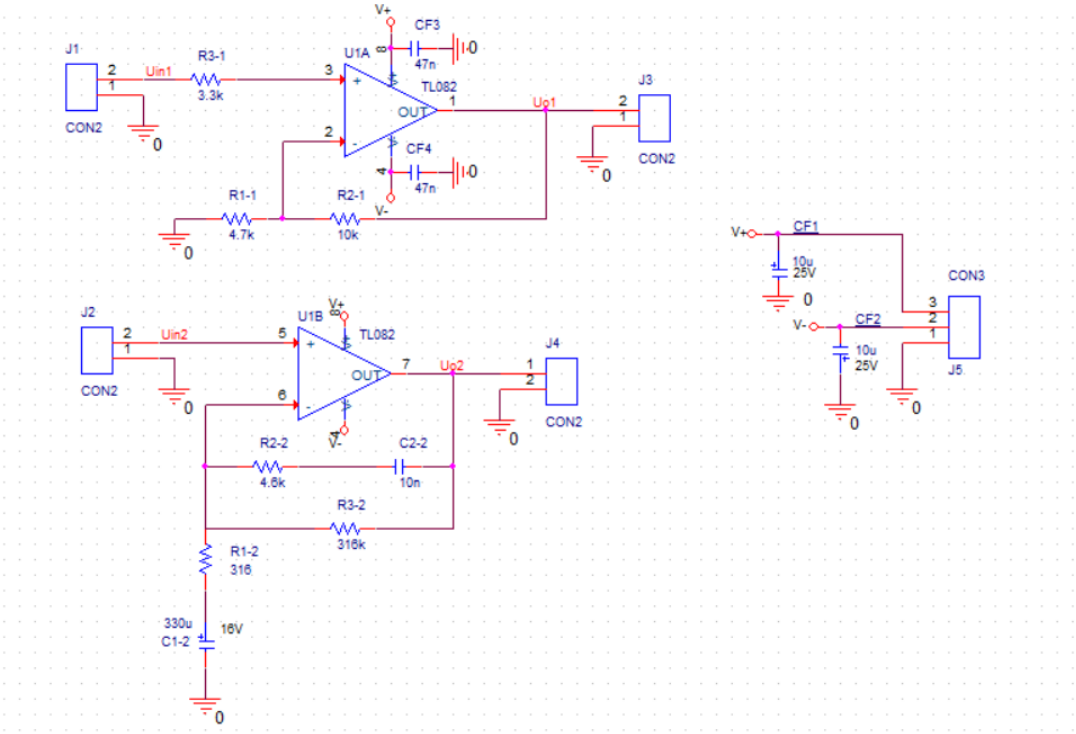


SET=0.5 la R2-4 și SET=0.5 la R4-4 – Nu se intampla nimic



Etapa 6

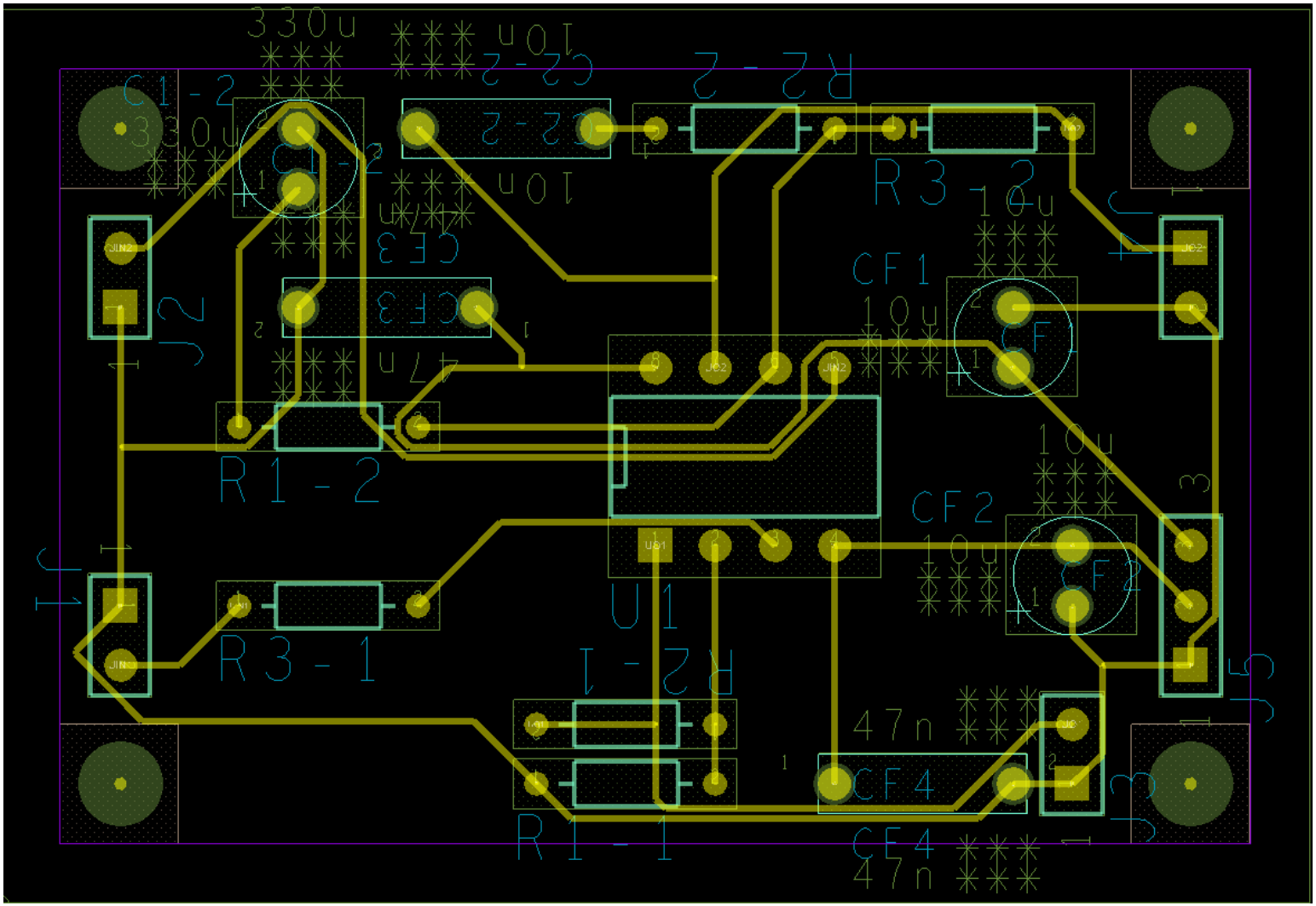
Circuitul final



BoM

Item	Quantity	Reference	Part	PCB Footprint
1	2	CF1,CF2	10u	CAP196
2	2	CF3,CF4	47n	CAP300
3	1	C1-2	330u	CAP196
4	1	C2-2	10n	CAP300
5	4	J1,J2,J3,J4	CON2	BLKCON100VHTM1SQW1002
6	1	J5	CON3	BLKCON100VHTM1SQW1003
7	1	R1-1	4.7k	AXRC05
8	1	R1-2	316	AXRC05
9	1	R2-1	10k	AXRC05
10	1	R2-2	4.6k	AXRC05
11	1	R3-1	3.3k	AXRC05
12	1	R3-2	316k	AXRC05
13	1	U1	TL082	DIP1008W300L450

PCB


C1-2
330u
C2-2
10n
R2-2
R3-2
10u
CF1
10u
CF
CE3
47n
R1-2
U1
CF2
10u
CF2
R3-1
R2-1
47n
CF4
R1-1
CE4
47n